Now take:

$$v = \underbrace{\Pi_h u}_{\text{projection of } u \in V \text{ into the subspace } V_h \subset V} \in V_h$$

$$\Pi_h u = \sum_{j=1}^{M} u(x_j)\phi_j(x)$$

interpolate $u(x)$ onto mesh using continuous piecewise linear fns.

Note: $\|u - \Pi_h u\|_E = \|(u - \Pi_h u)'\|_a \leq C\|hu''\|_a \leq C h_{max}\|u''\|_a$

(C: this comes from approx u' by a piecewise constant function.)

- C is a const that only depends on $a(x)$.

- Theorem: The FEM approx $U \in V_h$ satisfies $\boxed{\|u - U\|_E \leq \|u - v\|_E}$ $\forall v \in V_h \subset V$.

In particular, there exists a constant C, depending only on $a(x)$, such that $\|u - U\|_E \leq C\|hu''\|_a$.

Note: this shows that CG(1) is first-order accurate in the energy norm, provided that $u'' \in L^2(0,1)$.

(2) $L^2(0,1)$ norm:

$$\left\{ \begin{array}{l} -(au')' = f \text{ in } (0,1) \\ u(0) = u(1) = 0. \end{array} \right\}$$

Consider the following dual problem:

$$\left\{ \begin{array}{l} -(a\phi')' = e \\ \phi(0) = \phi(1) = 0 \end{array} \right\} \text{ where } e = u - U$$

Note: because the original 2-pt BVP is self-adjoint, primal operators and the dual operator are the same.

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L^2-norm: $\|e\|^2 = \int_0^1 e^2\,dx = -\int_0^1 e(a\phi')'\,dx = \int_0^1 ae'\phi'\,dx - [ae\phi']_0^1$

but $e(1) = e(0) = 0$

$\Rightarrow \|e\|^2 = \int_0^1 ae'\phi'\,dx$

Recall: Galerkin orthogonality: $\int_0^1 ae'v'\,dx = 0 \quad \forall\, v \in V_h$.

take $v = \pi_h\phi$

$$\|e\|^2 = \int_0^1 ae'\phi'\,dx - \int_0^1 ae'(\pi_h\phi)'\,dx = \int_0^1 ae'(\phi - \pi_h\phi)'\,dx$$

Cauchy ineq: $\|e\|^2 \leq \|e'\|_a \|(\phi - \pi_h\phi)'\|_a$

$$\Rightarrow \|e\|^2 \leq c_I h_{max} \|e'\|_a \underbrace{\|\phi''\|_a}_{\text{interp. error}}$$

but recall $\|e'\|_a = \|e\|_E \leq c_I h_{max} \|u''\|_a$

↳ from part (1).

$$\|e\|^2 \leq c_I^2 h_{max}^2 \|u''\|_a \|\phi''\|_a$$

$$\Rightarrow \|e\|^2 \leq c_I^2 h_{max}^2 \|u''\|_a \frac{\|\phi''\|_a}{\|e\|} \cdot \|e\|$$

$$\|e\| \leq c_I^2 h_{max}^2 \|u''\|_a \boxed{\frac{\|\phi''\|_a}{\|e\|}}$$

Let $S_a := \max_{g \neq 0} \frac{\|\psi''\|_a}{\|g\|} \leq C_S$ where $-(a\psi')' = g$

$\psi(0) = \psi(1) = 0$.

Note: if $a \equiv$ constant

$$-(a\psi')' = -a\psi'' = g \quad \Rightarrow \frac{\|\psi''\|}{\|g\|} = \frac{1}{a}$$

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Claim: S_a is bounded by a constant only depending on a. (C_S)

11/09/26:

Proof: $-a\psi'' = a'\psi' + \xi$

$$\|a\psi''\| \le \|\xi\| + \|a'\psi'\|$$
$$\le \|\xi\| + \|a'\|_\infty \cdot \|\psi'\|$$
$$\|a'\|_\infty = \max_{x\in[0,1]} |a'(x)|$$

Take BVP & multiply it by ψ and int-by-parts:

$$-\int_0^1 \psi (a\psi')' dx = \int_0^1 \xi\psi\, dx$$
$$\Rightarrow \int_0^1 a(\psi')^2 dx = \int_0^1 \xi\psi\, dx$$
$$\Rightarrow a_{\min}\int_0^1 (\psi')^2 dx \le \int_0^1 \xi\psi\, dx$$
$$\therefore \boxed{a_{\min}\|\psi'\|^2 \le \|\xi\|\cdot\|\psi\|}$$

Aside: Poincaré's inequality

Let $v\in V$, then $v(x) = \int_0^x v'(y)\,dy$

$$|v| \le \int_0^1 |v'|\,dy \le \sqrt{\int_0^1 (v')^2 dy}$$
$$\therefore \boxed{\int_0^1 v^2 dx \le \int_0^1 (v')^2 dy}$$
$$\|v\| \le \|v'\|$$

Energy norm: The energy norm comes straight from the weak form. If the weak form is $a(u,v) = (f,v)$ where $a(\cdot,\cdot)$ is bilinear form you get that from after integrating by parts, then the energy norm is:

$$\|v\|_E := \sqrt{a(v,v)}$$

In short, you plug the same fn into both slots of a and take sq. root.

Galerkin Orthogonality:

Exact soln: $a(u,v) = (f,v)\ \forall\ v\in V$

FEM soln: $a(U,v) = (f,v)\ \forall\ v\in V_h$

Because $V_h \subset V$, the first eqn holds for every $v\in V_h$. Subtracting the second from the first gives

$$a(u-U, v) = 0 \quad \forall\ v\in V_h$$

Meaning: the error $e = u-U$ is perpendicular to every function in V_h, where $a(\cdot,\cdot)$ plays the role of a dot product. So U is the projection of u onto V_h the same picture as dropping a $\perp$ar from a point onto a plane.

Cauchy-Schwarz inequality says an inner product is never bigger than the product of the two norms.

Vectors: $x\cdot y = |x||y|\cos\theta \le |x||y|$

Functions:

L^2 version: $\left|\int wv\,dx\right| \le \|w\|\,\|v\|$

Energy version: $|a(w,v)| \le \|w\|_E\,\|v\|_E$

$\|\cdot\|_a \to$ is just a weighted L^2 norm.

Back to proof:

$$a_{min}\|\psi'\|^2 \le \|e_y\|\,\|\psi\|$$

$$\le \|e_y\| \cdot \|\psi'\| \quad \text{(by Poincare's)}$$

$$\therefore \boxed{\|\psi'\| \le \frac{\|e_y\|}{a_{min}}}$$

$$\|a\psi''\| \le \|e_y\| + \|a'\|_\infty \|\psi'\|$$

$$\|a\psi''\| \le \|e_y\| + \|a'\|_\infty \frac{\|e_y\|}{a_{min}}$$

$$\|a\psi''\| \le \left(1 + \frac{\|a'\|_\infty}{a_{min}}\right)\|e_y\|$$

but $\sqrt{a_{min}}\,\|\psi''\|_a \le \|a\psi''\|$

$$\left(\|\psi''\|_a = \sqrt{\int_0^1 a(\psi'')^2 dx}\,, \text{ but } \|a\psi''\| = \sqrt{\int_0^1 a^2(\psi'')^2 dx}\right)$$

$$\therefore \|\psi''\|_a \le \underbrace{\left(\frac{1 + \frac{\|a'\|_\infty}{a_{min}}}{\sqrt{a_{min}}}\right)}_{C_s}\|e_y\|$$

$$\therefore \frac{\|\psi''\|_a}{\|e_y\|} \le C_s \qquad \blacksquare$$

$$\therefore \|e\| \le \underbrace{C_I^2 C_s}_{C}\|h^2 u''\|_a$$

—or—

$$\boxed{\|e\| \le \underbrace{C_I^2 C_s}_{C} h_{max}^2 \|u''\|_a}$$

$\therefore$ assuming $\|u''\|_a$ is bounded, then $\|e\| = O(h^2)$.

$\Rightarrow$ a posteriori error:

(1) energy norm:

$$\|e\|_E^2 = \|e'\|_a^2 = \int_0^1 a(e')^2 dx$$

$$= \int_0^1 au'e'dx - \int_0^1 aU'e'dx$$

$$= \int_0^1 fe\,dx - \int_0^1 aU'e'dx$$

Subtract from this:

$$\Rightarrow \int_0^1 fv\,dx - \int_0^1 aU'v'dx = 0 \quad \text{w/ } v = \pi_h e \in V_h$$

$$\therefore \|e\|_E^2 = \int_0^1 f(e-\pi_h e)dx - \int_0^1 aU'(e-\pi_h e)'dx$$

$$\Rightarrow -\int_0^1 aU'(e-\pi_h e)'dx = -\sum_{j=1}^{M+1}\int_{x_{j-1}}^{x_j} aU'(e-\pi_h e)'dx = \sum_{j=1}^{M+1}\int_{x_{j-1}}^{x_j}(aU')'(e-\pi_h e)\,dx - \sum_{j=1}^{M+1}\left[aU'(e-\pi_h e)\right]_{x_{j-1}}^{x_j}$$

but $e-\pi_h e\big|_{x_j} = 0 \quad \forall j$

$$\therefore \|e\|_E^2 = \sum_{j=1}^{M+1}\int_{x_{j-1}}^{x_j} R(U)(e-\pi_h e)dx$$

where the element residual is $\boxed{R(U) := f + (aU')'}$

- Cauchy Inequality:

$$\|e\|_E^2 \le \|h\,R(U)\|_{a^{-1}} \cdot \|h^{-1}(e-\pi_h e)\|_a$$

but from interpolation:

$$\|h^{-1}\underbrace{(e-\pi_h e)}_{O(h^2)}\|_a \le C\underbrace{\|e'\|_a}_{O(h)}$$

- **Theorem**: There exists a constant 'c' depending only on 'a' such that the CG(1) FEM approximation.

$$\|e\|_E = \|(u-U)'\|_a \leq c\|hR(U)\|_{a^{-1}}$$

Note: Roughly speaking: $R(U) = O(1) \quad \therefore \|e\|_E = O(h)$

Note: can do adaptive error control by enforcing $c\|hR(U)\|_{a^{-1}} \leq \varepsilon$ (user-specified tolerance).

- In practice, can do this as follows:

(1) on some initial coarse mesh compute the FEM soln.

(2) Calculate on each element the "residual":-

$$E_j^2 = (x_j - x_{j-1})^2 \int_{x_{j-1}}^{x_j} \frac{1}{a(x)} R^2(U)\,dx$$

where $R(U) = f + (aU')'$

want $E^2 = \sum_{j=1}^{M+1} E_j^2 \leq \frac{\varepsilon^2}{c^2}$

If $E_j^2 \leq \frac{\varepsilon^2}{c^2(M+1)}$ then move on to next element.

If $E_j^2 > \frac{\varepsilon^2}{c^2(M+1)}$ then subdivide the element (typically into 2 element).

(3) Recalculate solution on new mesh, then repeat steps (2)+(3) until no subdivision is required.

Note: improved efficiency if c is known. In practice, some calibration on known solutions is used.

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(2) L^2 norm:

Theorem: (L^2 norm a posteriori error estimate).

$$\|e\| = \|u-U\| \le C_S C_I \|h^2 R(U)\|$$

where $\|h^{-2}(\phi - \pi_h\phi)\| \le C_I \|\phi''\|$

$$C_S = \max_{g \neq 0} \frac{\|\phi''\|}{\|g\|} \quad \text{where} \quad -(a\phi')' = g \quad \phi(0)=\phi(1)=0.$$

$$\begin{cases} -(a\phi')' = e \\ \phi(0)=\phi(1)=0 \end{cases}$$ "dual" problem.

Proof:

$$\|e\|^2 = \int_0^1 e^2 dx = -\int_0^1 e(a\phi')' dx$$

$$\|e\|^2 = -\int_0^1 u(a\phi')' dx + \int_0^1 U(a\phi')' dx$$

$$= \int_0^1 au'\phi' dx - \int_0^1 aU'\phi' dx$$

$$= \int_0^1 f\phi\, dx - \int_0^1 aU'\phi' dx$$

then using FEM with $v = \pi_h\phi$:

$$\int_0^1 f\pi_h\phi\, dx - \int_0^1 aU'(\pi_h\phi)' dx = 0$$

$$\|e\|^2 = \int_0^1 f(\phi - \pi_h\phi) dx - \sum_{j=1}^{M+1} \int_{x_{j-1}}^{x_j} aU'(\phi - \pi_h\phi)' dx$$

reverse int-by-parts:

$$\|e\|^2 = \sum_{j=1}^{M+1} \int_{x_{j-1}}^{x_j} R(U)(\phi - \pi_h\phi) dx \quad \text{where } R(U) = f + (aU')'$$

Cauchy-Ineq: $\|e\|^2 \le \|h^2 R(U)\| \, \|h^{-2}(\phi - \pi_h\phi)\|$

interpolation: $\|h^{-2}(\phi - \pi_h\phi)\| \le C_I \|\phi''\|$

$$\therefore \|e\|^2 \le C_I \|h^2 R(U)\| \cdot \frac{\|\phi''\|}{\|e\|} \cdot \|e\| \qquad \left(\frac{\|\phi''\|}{\|e\|} \le C_S\right)$$

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$$\therefore \boxed{\|e\| \le c_I c_S \|h^2 R(U)\|}$$

Note: modify step 2 in algorithm:

(2) Calculate on each element:

$$\boxed{E_j^2 := (x_j - x_{j-1})^4 \int_{x_{j-1}}^{x_j} R^2(U(x))\,dx}$$

If $E_j^2 \le \frac{\varepsilon^2}{c^2(M+1)}$ move to next element; else refine.

➡ CG(2) method: $\begin{cases} -(au')' = f & \text{on } (0,1) \\ u(0) = u(1) = 0 \end{cases}$

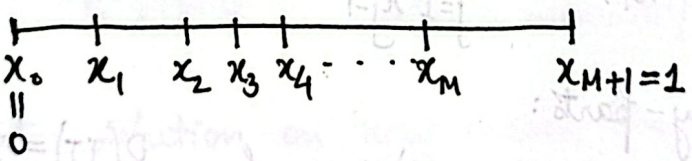
$$\int_0^1 a u' v' dx = \int_0^1 f v\,dx \quad \forall\, v \in V$$

$$V = \left\{ v : \int_0^1 (v^2 + (v')^2)\,dx < \infty \,,\ v(0) = v(1) = 0 \right\}$$

For CG(2):

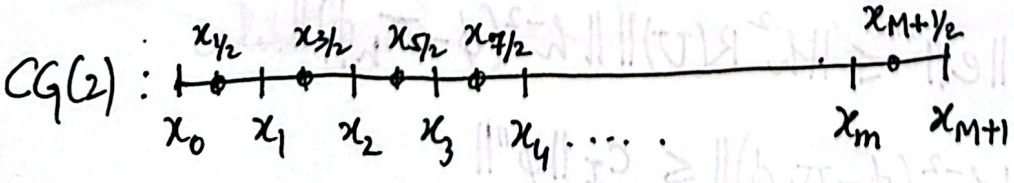
$$V_h = \left\{ v \in V : v|_{[x_{j-1}, x_j]} = a_j + b_j x + c_j x^2 \right\}$$

Key: need to find a suitable basis for V_h.

Mesh: T_h: 

Simplest way to write CG(2) is to use Lagrangian polys & to do this, introduce a "submesh".

[Need not be uniform]

CG(2): 

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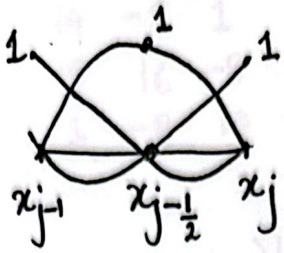
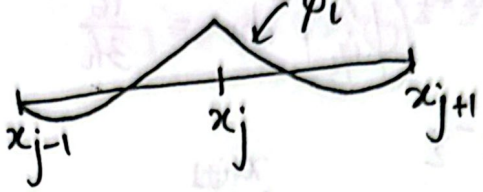
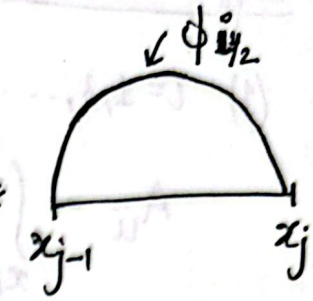
For a given element, there are 3 Lagrange polys:



2-basis functions:





$$\phi_i(x) = \begin{cases} \dfrac{2(x-x_{i+\frac{1}{2}})(x-x_{i+1})}{(x_{i+1}-x_i)^2} & x_i \le x \le x_{i+1} \\ \dfrac{2(x-x_{i-\frac{1}{2}})(x-x_{i-1})}{(x_i-x_{i-1})^2} & x_{i-1} \le x \le x_i \\ 0 & \text{otherwise} \end{cases}$$

$$\phi_{i-\frac{1}{2}}(x) = \begin{cases} \dfrac{4(x_i-x)(x-x_{i-1})}{(x_i-x_{i-1})^2} & x_{i-1} \le x \le x_i \\ 0 & \text{otherwise} \end{cases}$$

$$\therefore V_h = \text{Span}\left\{\phi_{\frac{1}{2}}, \phi_1, \phi_{\frac{3}{2}}, \phi_2, \dots, \phi_M, \phi_{M+\frac{1}{2}}\right\}$$

$$\therefore \text{if } U \in V_h \text{ then } U(x) = \sum_{j=1}^{2M+1} \xi_j \phi_{j/2}(x)$$

Note: for any function $v \in V$: $\Pi_h v = \sum_{K=1}^{2M+1} v(x_{\frac{K}{2}})\phi_{\frac{K}{2}}(x)$

test functions: $v = \phi_{i/2}(x)$ for $i = 1, 2, \dots, 2M+1$.

⇒ CG(2) Var. Prob:

$$\sum_{i=1}^{2M+1} \xi_j \int_0^1 a(x)\, \phi'_{i/2} \phi'_{j/2}\, dx = \int_0^1 f \phi_{i/2}\, dx$$

∵ $(2M+1)\times(2M+1)$ symmetric pos. definite system, but not tridiagonal.

ex: consider $a \equiv 1$ & $h_j = h \ \forall j$

Stiffness matrix:

(1) $i = 1, 3, \ldots, 2M+1$

$$A_{ii} = \int_{x_{\frac{i}{2}-\frac{1}{2}}}^{x_{\frac{i}{2}+\frac{1}{2}}} \left(\phi'_{i/2}\right)^2 dx = \frac{16}{3h}$$

$$A_{(i-1,i)} = A_{(i,i-1)} = \int_{x_{\frac{i-1}{2}}}^{x_{\frac{i+1}{2}}} \phi'_{i/2}\, \phi'_{\frac{i-1}{2}}\, dx = \frac{-8}{3h}$$

(2) $i = 2, 4, 6, \ldots, 2M$

$$A_{ii} = \int_{x_{\frac{i-1}{2}}}^{x_{\frac{i+1}{2}}} \left[\phi'_{i/2}\right]^2 dx = \frac{14}{3h}$$

$$A_{(i,i-1)} = A_{(i-1,i)} = \int_{x_{\frac{i-1}{2}}}^{x_{\frac{i+1}{2}}} \phi'_{i/2}\, \phi'_{\frac{i}{2}-\frac{1}{2}}\, dx = \frac{-8}{3h}$$

$$A_{(i,i-2)} = A_{(i-2,i)} = \int_{x_{\frac{i-1}{2}}}^{x_{i/2}} \phi'_{i/2}\, \phi'_{\frac{i}{2}-1}\, dx = \frac{1}{3h}$$

RHS vector:

$$f \approx \tilde{\Pi}_h f = \sum_{j=0}^{2M+2} f(x_{j/2})\, \phi_{j/2}(x)$$

(1) $i = 1, 3, \ldots, 2M+1$

$$f_i = \sum_{j=1}^{2M+1} f(x_{j/2}) \int_{x_{\frac{i-1}{2}}}^{x_{\frac{i+1}{2}}} \phi_{j/2}\, \phi_{i/2}\, dx = \frac{h}{15}\left[f(x_{\frac{i-1}{2}}) + 8 f(x_{i/2}) + f(x_{\frac{i+1}{2}})\right]$$

(2) $i = 2, 4, \ldots, 2M$

$$f_i = \frac{h}{30}\left[-f(x_{\frac{i-2}{2}}) + 2f(x_{\frac{i-1}{2}}) + 8f(x_{i/2}) + 2f(x_{\frac{i+1}{2}}) - f(x_{\frac{i+2}{2}})\right]$$

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$$\therefore \underline{\underline{A}} = \frac{1}{3h}\begin{bmatrix} 16 & -8 & & & & & & \\ -8 & 14 & -8 & 1 & & & & \\ & -8 & 16 & -8 & & & & \\ & 1 & -8 & 14 & -8 & 1 & & \\ & & \ddots & \ddots & \ddots & \ddots & \ddots & \\ & & & 1 & -8 & 14 & -8 & \\ & & & & & -8 & 16 & \end{bmatrix}$$

$$\underline{\underline{A}}\,\underline{x} = \underline{b}$$

Note: energy norm error:

$$\|u-U\|_E \le \|u-v\|_E \quad \forall\, v \in V_h$$

take $v = \Pi_h u$

$$\therefore \boxed{\|u-U\|_E \le c\|h^2 u'''\|.}$$

Note: L^2 error

$$\boxed{\|u-U\| \le c\|h^3 u'''\|.}$$

⇒ **Beam equation:**

$$\begin{cases} u^{(iv)} = f \quad \text{on } 0<x<1 \\ u(0) = u'(0) = u(1) = u'(1) = 0 \end{cases}$$

Variation Problem:

$$\int_0^1 v u^{(iv)}\,dx = \int_0^1 f v\,dx$$

$$= -\int_0^1 v' u'''\,dx + [v u''']_0^1 = \int_0^1 f v\,dx$$

let $v(0) = v(1) = 0$

$$\int_0^1 v'' u''\,dx - [v' u'']_0^1 = \int_0^1 f v\,dx$$

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Let $v'(0) = v'(1) = 0$

$\therefore \int_0^1 u''v''\,dx = \int_0^1 fv\,dx$

$V = \left\{ v : \int_0^1 [v^2 + (v')^2 + (v'')^2]\,dx < \infty \quad v(0) = v(1) = v'(0) = v'(1) = 0 \right\}$

functions in V are $C^1[0,1]$.

- $\left\{ \text{Find } u \in V \text{ s.t } \int_0^1 u''v''\,dx = \int_0^1 fv\,dx \quad \forall v \in V. \right\}$ Variational problem.

⇒ Energy Minimization:

Define: $F(w) = \frac{1}{2}\int_0^1 (w'')^2\,dx - \int_0^1 fw\,dx$

(Min Prob): Find $u \in V$ s.t, $F(u) \leq F(w)\ \forall w \in V$.

Claim: (Var Prob) ⇔ (Min Prob)

(⇒) Assume $u \in V$ satisfies Var. Prob:

Let $w = u + v \in V$

$F(w) = F(u+v) = \frac{1}{2}\int_0^1 (u'')^2\,dx + \frac{1}{2}\int_0^1 (v'')^2\,dx + \int_0^1 u''v''\,dx - \int_0^1 fu\,dx - \int_0^1 fv\,dx$

$F(w) = F(u) + \frac{1}{2}\int_0^1 (v'')^2\,dx$

$\therefore F(w) \geq F(u)$

Note: $F(w) = F(u) \Rightarrow v'' \equiv 0 \Rightarrow v \equiv 0 \Rightarrow u \equiv w$.

due to B.Cs

→ This statement is true because it is C^1 for and for fn to be C^1, it should be continuous (value match) and it's first derivative should be continuous (derivative value match) at each point in the domain including element interfaces.

⇒ CG(3) Method:

$V_h \subseteq V$.

On each element: $[x_{j-1}, x_j]$

- Need a poly that matches value & derivative to its neighbors at x_{j-1} & x_j ⇒ 4 conditions.

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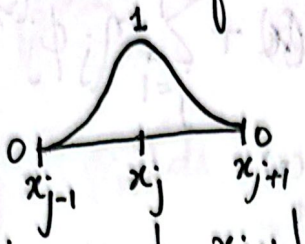
∴ need a cubic poly (4 coefficients).

→ These basis fns are also Hermite polynomials where you interpolate fn & fn derivate values at the element boundaries. The prev. one was Lagrange Polynomials.

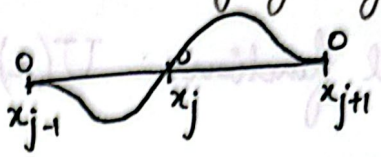
$$V_h = \{ v \in V : v|_{[x_{j-1}, x_j]} = a_j + b_j x + c_j x^2 + d_j x^3 \}$$

Key: Find a basis for V_h.

$\phi_j^{(i)}(x)$:



$\phi_j^{(2)}(x)$:



	x_{j-1}	x_j	x_{j+1}
$\phi_j^{(1)}$:	0	1	0
$(\phi_j^{(1)})'$:	0	6	0
$\phi_j^{(2)}$:	0	0	0
$(\phi_j^{(2)})'$:	0	1	0

$$\phi_j^{(1)} = \begin{cases} 3\left(\frac{x-x_{j-1}}{x_j-x_{j-1}}\right)^2 - 2\left(\frac{x-x_{j-1}}{x_j-x_{j-1}}\right)^3 & x_{j-1} \le x \le x_j \\ 3\left(\frac{x-x_{j+1}}{x_j-x_{j+1}}\right)^2 - 2\left(\frac{x-x_{j+1}}{x_j-x_{j+1}}\right)^3 & x_j \le x \le x_{j+1} \\ 0 \end{cases}$$

$$\phi_j^{(2)} = \begin{cases} -\frac{1}{h_j}(x-x_{j-1})^2 + \frac{1}{h_j^2}(x-x_{j-1})^3 & x_{j-1} \le x \le x_j \\ \frac{1}{h_{j+1}}(x-x_{j+1})^2 + \frac{1}{h_{j+1}^2}(x-x_{j+1})^3 & x_j \le x \le x_{j+1} \\ 0 & \text{otherwise} \end{cases}$$

CG(3) FEM:

Find $U \in V_h$ s.t $\int_0^1 U'' v'' dx = \int_0^1 f v dx \quad \forall v \in V_h$

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$V_h = \text{Span}\{\phi_1^{(1)}, \phi_2^{(1)}, \dots, \phi_M^{(1)}, \phi_1^{(2)}, \phi_2^{(2)}, \dots, \phi_M^{(2)}\}$

$\dim V_h = 2M$

trial functions: $U(x) = \sum_{j=1}^{M} \xi_j \phi_j^{(1)}(x) + \sum_{j=1}^{M} \eta_j \phi_j^{(2)}(x)$

test functions: $v = \phi_1^{(1)}, \dots, \phi_M^{(2)}$

Also, for any $u \in V_h$:

$$\pi_h u = \sum_{j=1}^{M} u(x_j)\phi_j^{(1)}(x) + \sum_{j=1}^{M} u'(x_j)\phi_j^{(2)}(x) \quad \text{where } \pi_h u \in V_h$$

⇒ Interpolation errors:

$$\|u - \pi_h u\|_{L^2(0,1)} \le c_1 \|h^4 u^{(4)}\|_{L^2(0,1)}$$

$$\|(u - \pi_h u)'\|_{L^2(0,1)} \le c_2 \|h^3 u^{(4)}\|_{L^2(0,1)}$$

$$\|(u - \pi_h u)''\|_{L^2(0,1)} \le c_3 \|h^2 u^{(4)}\|_{L^2(0,1)}$$

ex: (uniform mesh) $h_j \equiv h \ \forall j$

$$\int_0^1 U''v''dx = \int_0^1 fv dx$$

$$v = \phi_1^{(1)}, \phi_2^{(1)}, \dots, \phi_M^{(1)}, \phi_1^{(2)}, \dots, \phi_M^{(2)}$$

Stiffness matrix: $\int_0^1 U''v''dx = \sum_{j=1}^{M} \xi_j \int_0^1 (\phi_j^{(1)})''(\phi_k^{(1)})''dx + \sum \eta_j \int_0^1 \phi_j^{(2)''}\phi_k^{1''}dx$

$k = 1, 2, \dots, M.$

Transpose ↓ each other

$$\& \sum_{j=1}^{M} \xi_j \int_0^1 (\phi_j^{(1)})''(\phi_k^{(2)})''dx + \sum_{j=1}^{M} \eta_j \int_0^1 (\phi_j^{(2)})''(\phi_k^{(2)})''dx$$

$k = 1, 2, \dots M$

$$\begin{bmatrix} \underline{\underline{A}} & \vdots & \underline{\underline{C}} \\ \cdots & \cdots & \cdots \\ -\underline{\underline{C}} & \vdots & \underline{\underline{B}} \end{bmatrix} \in \mathbb{R}^{2M \times 2M}$$

$$\underline{\underline{A}}, \underline{\underline{B}}, \underline{\underline{C}} \in \mathbb{R}^{M \times M}$$

$$\rightarrow A_{jk} = \int_0^1 (\phi_j^{(1)})'' (\phi_k^{(1)})'' dx$$

$$\underline{\underline{A}} = \frac{12}{h^3} \begin{bmatrix} 2 & -1 & & & \\ -1 & 2 & -1 & & \\ & \ddots & \ddots & \ddots & \\ & & \ddots & \ddots & -1 \\ & & & -1 & 2 \end{bmatrix}$$

$\therefore$ M×M, tridiagonal & symmetric

$$\rightarrow B_{jk} = \int_0^1 (\phi_j^{(2)})'' (\phi_k^{(2)})'' dx$$

$$\underline{\underline{B}} = \frac{2}{h} \begin{bmatrix} 4 & 1 & & & \\ 1 & 4 & 1 & & \\ & \ddots & \ddots & \ddots & \\ & & \ddots & \ddots & 1 \\ & & & 1 & 4 \end{bmatrix}$$

$\therefore$ M×M, tridiagonal, symmetric

$$\rightarrow C_{jk} = \int_0^1 (\phi_j^{(1)})'' (\phi_k^{(2)})'' dx$$

$$\underline{\underline{C}} = \frac{6}{h^2} \begin{bmatrix} 0 & 1 & & & \\ -1 & \ddots & \ddots & & \\ & \ddots & \ddots & \ddots & \\ & & \ddots & \ddots & 1 \\ & & & -1 & 0 \end{bmatrix}$$

$\therefore$ M×M, tridiagonal, $\underline{\underline{C}}^T = -\underline{\underline{C}}$

$\Rightarrow$ is anti-symmetric

$$\therefore \underline{\underline{S}} = \begin{bmatrix} \underline{\underline{A}} & \vdots & \underline{\underline{C}} \\ \cdots & \cdots & \cdots \\ \underline{\underline{C}}^T & \vdots & \underline{\underline{B}} \end{bmatrix}$$

$$\therefore \underline{\underline{S}}^T = \underline{\underline{S}}$$

$\therefore$ $\underline{\underline{S}}$ is 2M×2M, symmetric

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RHS:

$$f \to \tilde{\Pi}_h f = \sum_{j=0}^{M+1} f(x_j)\,\phi_j^{(1)}(x) + \sum_{j=0}^{M+1} f'(x_j)\,\phi_j^{(2)}(x)$$

$$\int_0^1 f v\,dx \to \int_0^1 (\tilde{\Pi}_h f)\cdot v\,dx$$

$$f_j^{(1)} = \frac{h}{70}\left[9f(x_{j-1}) + 52f(x_j) + 9f(x_{j+1})\right] + \frac{13h^2}{420}\left[f'(x_{j-1}) - f'(x_{j+1})\right]$$

$$j = 1, 2, \cdots, M$$

$$f_j^{(2)} = \frac{13h^2}{420}\left[f(x_{j+1}) - f(x_{j-1})\right] - \frac{h^3}{420}\left[3f'(x_{j-1}) - 8f'(x_j) + 3f'(x_{j+1})\right]$$

$$j = 1, 2, \cdots, M$$

∴ final 2M×2M system:

$$\begin{bmatrix} \underline{\underline{A}} & \underline{\underline{C}} \\ -\underline{\underline{C}} & \underline{\underline{B}} \end{bmatrix} \begin{bmatrix} \underline{x} \\ \underline{y} \end{bmatrix} = \begin{bmatrix} \underline{f}^{(1)} \\ \underline{f}^{(2)} \end{bmatrix}$$

2M×2M, Symmetric, Positive definite.

⇒ A-priori error estimates:

(1) energy norm: $\|e\|_E^2 = \int_0^1 (e'')^2\,dx$

Find $u \in V$: $\int_0^1 u''v''dx = \int_0^1 fv dx \quad \forall\ v \in V.$

Find $U \in V_h \subset V$: $\int_0^1 U''v''dx = \int_0^1 fv dx$

$\therefore$ Galerkin orthogonality: $\int_0^1 (u-U)''v''dx = 0 \quad \forall\ v \in V_h$

$$\|u-U\|_E^2 = \int_0^1 (u-U)''(u-U)''dx = \int_0^1 (u-U)''(u-v+v-U)''dx$$

$$\|u-U\|_E^2 = \int_0^1 (u-U)''(u-v)''dx + \int_0^1 (u-U)''(v-U)''dx$$

$(v-U)'' \to 0$ for some $v \in V_h$.

$$\|u-U\|_E^2 = \int_0^1 (u-U)''(u-v)''dx$$

by cauchy ineq: $\|u-U\|_E^2 \le \|u-U\|_E \cdot \|u-v\|_E$

$$\boxed{\|u-U\|_E \le \|u-v\|_E \quad \forall\ v \in V_h}$$

$\therefore U$ is best soln from V_h in the energy norm.

let $v = \Pi_h u$.

$$\|u-U\|_E \le \|u-\Pi_h u\|_E$$

$$\boxed{\|u-U\|_E \le C \|h^2 u^{(4)}\|_{L^2(0,1)}}$$

$$\|h^2 u^{(4)}\|_{L^2(0,1)}^2 = \sum_{j=1}^{M+1} \int_{x_{j-1}}^{x_j} h_j^2 (u^{(4)})^2 dx$$

(2) L^2 error estimate:

Claim: $\mathcal{L}(u) = u^{(iv)}$ w/ $u(0) = u(1) = u''(0) = u'(1) = 0$ is self-adjoint.

Proof: $\int_0^1 v\mathcal{L}(u)\,dx$ where $u \& v$ satisfy BCs

$$\Rightarrow \int_0^1 v\,u^{(iv)}\,dx = -\int_0^1 v'u'''\,dx + [vu''']_0^1$$

$$\Rightarrow \int_0^1 v''u''\,dx + [vu''' - v'u'']_0^1$$

$$\Rightarrow -\int_0^1 v'''u'\,dx + [vu''' - v'u'' + v''u']_0^1$$

$$\Rightarrow \int_0^1 v^{(4)}u\,dx + [vu''' - v'u'' + v''u' - v'''u]_0^1$$

$$\therefore \int_0^1 v\,u^{(4)}\,dx = \int_0^1 u\,v^{(4)}\,dx$$

Primal Problem:

$$\left\{ \begin{array}{l} u^{(4)} = f \\ u(0) = u(1) = u'(0) = u'(1) = 0 \end{array} \right\}$$

Dual Problem:

$$\left\{ \begin{array}{l} \phi^{(4)} = e \\ \phi(0) = \phi(1) = \phi''(0) = \phi''(1) \end{array} \right\}$$

$$\|e\|^2 = \int_0^1 e^2\,dx = \int_0^1 e\,\phi^{(4)}\,dx = \int_0^1 e''\,\phi''\,dx$$

Boundary terms vanish: $e(0) = e(1) = 0$, $e'(0) = e'(1) = 0$ | $\because e = u - U$ & both satisfy B.Cs.

Galerkin Orthogonality: $\int_0^1 e''(\pi_h\phi)''\,dx = 0.$

$$\|e\|^2 = \int_0^1 e''(\phi - \pi_h\phi)''\,dx$$

$$\leq \|e''\| \cdot \|(\phi - \pi_h\phi)''\|$$

$$\|e\|_E$$

$$\|e\|^2 \leq c\|h^2\phi^{(4)}\| \cdot c\|h^2u^{(4)}\|$$

Let $h_j \leq h_{max}\ \forall j$

$$\|e\|^2 \leq c^2 h_{max}^2 \|\phi^{(4)}\| \cdot \|u^{(4)}\|$$

$$\|e\|^2 \leq c^2 h_{max}^2 \cdot \frac{\|\phi^{(4)}\|}{\|e\|} \cdot \|e\| \cdot \|u^{(4)}\| \quad \left(\frac{\|\phi^{(4)}\|}{\|e\|} = 1\right)$$

$$\Rightarrow \boxed{\|e\| \leq c^2 h_{max}^4 \|u^{(4)}\|}$$

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Abstract Formulation :

So far : 2 examples

(1) $-(au')' = f$ in $(0,1)$

$u(0) = u(1) = 0$

$V = \{v : \int_0^1 [v^2 + (v')^2]\,dx < \infty,\ v(0) = v(1) = 0\}$

$B(u,v) = \int_0^1 au'v'\,dx$

$L(v) = \int_0^1 fv\,dx$

Var Prob : Find $u \in V$ s.t $B(u,v) = L(v)\ \forall\ v \in V$

$F(w) = \frac{1}{2} B(w,w) - L(w)$

Energy Min Prob : Find $u \in V$ s.t $F(u) \le F(w)\ \forall\ w \in V$.

(2) Beam eqn :

$V = \{v : \int_0^1 [v^2 + (v')^2 + (v'')^2]\,dx < \infty,\ v(0) = v(1) = v'(0) = v'(1) = 0\}$

$B(u,v) = \int_0^1 u''v''\,dx$

and $L(v) = \int_0^1 fv\,dx$.

Note : we seek a ~~conforming~~ conforming FEM : $V_h \subset V$.

(non-conforming FEM : $V_h \not\subset V$.)

Abstract formulation : Consider the domain $\Omega \subset \mathbb{R}^d$.

Define (derivatives via a multi-index)

Let $\underline{\alpha} = (\alpha_1, \alpha_2, \cdots, \alpha_d)$ be a d-tuple, where $\alpha_1, \alpha_2, \cdots, \alpha_d$ are non-negative integers

We refer to $\underline{\alpha}$ as a multi-index ; the length of α is $|\alpha| = \alpha_1 + \alpha_2 + \cdots + \alpha_d$.

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The derivative of v is:

$$D^{\alpha} v = \frac{\partial^{|\alpha|} v}{\partial x_1^{\alpha_1} \partial x_2^{\alpha_2} \cdots \partial x_d^{\alpha_d}}$$

Ex: If $d = 2$ and $|\alpha| = 2$

$$D^{(2,0)} v = \frac{\partial^2 v}{\partial x_1^2}$$

$$D^{(0,2)} v = \frac{\partial^2 v}{\partial x_2^2}$$

$$D^{(1,1)} v = \frac{\partial^2 v}{\partial x_1 \partial x_2}$$

Define: $L^1(\Omega)$ norm:

$$\|u\|_{L^1(\Omega)} := \int_{\Omega} |u| \, dx$$

Defn: $L^1_{loc}(\Omega) := \{ v : v \in L^1(K)$ for any compact subset K inside $\Omega \}$

Defn: (Sobolev space)

$$W^{r,q}(\Omega) = \{ v \in L^1_{loc}(\Omega) : \|v\|_{W^{r,q}(\Omega)} < \infty \}$$

where

$$\|v\|_{W^{r,q}(\Omega)} := \left[\sum_{|\alpha| \le r} \|D^{\alpha} v\|^q_{L^q(\Omega)} \right]^{1/q}$$

$$1 \le q \le \infty$$

$$r \ge 0$$

and

$$\|v\|^q_{L^q(\Omega)} = \int_{\Omega} |v|^q \, dx$$

Defn: (Sobolev w/ Homogeneous B.Cs)

$$W^{r,q}_0(\Omega) = \{ v \in W^{r,q}(\Omega) : D^{\alpha} v \big|_{\partial\Omega} = 0 \text{ for } |\alpha| < r \}$$

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Defn: (Hilbert Space): When $q=2$ we write $W^{r,2}(\Omega) = H^r(\Omega)$. This is called a Hilbert space.

Norm: $\|v\|_{H^r(\Omega)} = \sqrt{(v,v)_{H^r(\Omega)}}$

Inner Product: $(v,w)_{H^r(\Omega)} := \sum_{|\alpha|\le r} \int_\Omega (D^\alpha v)(D^\alpha w)\, d\underline{x}$

Note: $r=0$ also makes sense: $H^0(\Omega) = L^2(\Omega)$.

Defn: (Semi-norms): $|v|_{H^r(\Omega)} := \sqrt{\sum_{|\alpha|=r} (D^\alpha v)^2 d\underline{x}}$

Note: for the remainder of discussion, let V be a Hilbert space w/ inner product $(\cdot,\cdot)_V$ and corresponding norm $\|\cdot\|_V$. $r=?$

Defn: A mapping $L: V \to \mathbb{R}$ is called a linear functional if $L(\alpha u + \beta v) = \alpha L(u) + \beta L(v)$ for all $\alpha, \beta \in \mathbb{R}$ and all $u, v \in V$. We say that L is bounded in the norm $\|\cdot\|_V$ if there exists a constant $\tilde{L} > 0$ such that

$$|L(v)| \le \tilde{L} \|v\|_V \quad \forall\, v \in V.$$

Note: A bounded linear functional is continous:

$$L(v) - L(w) = L(v-w)$$

$$\Rightarrow |L(v) - L(w)| = |L(v-w)| \le \tilde{L} \|v-w\|_V \quad ? \quad \forall\, v, w \in V.$$

The converse is also true: a continous linear functional is bounded.

Defn: The set of bounded linear functionals on V is termed the dual space of V and is denoted by V'. V' is equipped w/ a norm.

$$\|L\|_{V'} = \sup_{\substack{v \in V \\ v \neq 0}} \frac{L(v)}{\|v\|_V} \le \tilde{L}$$

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ex: (Dual space of $L^q(\Omega)$) Consider the linear functional

$$L(v) = \int_\Omega f(x)\, v(x)\, dx \quad \text{for all } f \in L^{q'}(\Omega) \text{ where } \frac{1}{q} + \frac{1}{q'} = 1.$$

Hölder's Inequality: $\|uv\|_{L^1(\Omega)} \le \|u\|_{L^q(\Omega)} \cdot \|v\|_{L^{q'}(\Omega)}$

$\forall\ u \in L^q(\Omega)$ & $v \in L^{q'}(\Omega)$

$\therefore |L(v)| \le \|f\|_{L^{q'}(\Omega)} \|v\|_{L^q(\Omega)}$

$\therefore$ every function $f \in L^{q'}(\Omega)$ can be viewed as a linear bounded functional on $L^q(\Omega)$

Every bounded linear functional on $L^q(\Omega)$ is of this form,

$\therefore$ $L^{q'}(\Omega)$ can be viewed as the dual space of $L^q(\Omega)$.

Thm: (Riesz Representation Thm): Let H be a Hilbert space w/ scalar product $(\cdot,\cdot)_H$. Then for any continous linear functional, $L: H \to \mathbb{R}$, there exists a unique $u \in H$ such that

$L(v) = (u,v)_H \quad \forall\ v \in H.$

Abstract Variational Problems:

Defn: (Bilinear form) Let V be a Hilbert space w/ scalar product $(\cdot,\cdot)_V$. Then $B(\cdot,\cdot): V \times V \to \mathbb{R}$ is a bilinear form if

$B(u, \alpha v + \beta w) = \alpha B(u,v) + \beta B(u,w)$

& $B(\alpha u + \beta v, w) = \alpha B(u,w) + \beta B(v,w)$ for all $\alpha, \beta \in \mathbb{R}$ and for all $u, v, w \in V$.

Defn: (Symmetric) The bilinear form $B(\cdot,\cdot): V \times V \to \mathbb{R}$ is symmetric if $B(u,v) = B(v,u)$ $\forall\ u, v \in V$. (S)

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Defn: (Continous / Bounded) B is continous (aka bounded) if there exists a constant $\beta^* > 0$ such that:

$$B(u,v) \leq |B(u,v)| \leq \beta^* \|u\|_V \|v\|_V \cdot \forall\, u,v \in V. \quad (CB).$$

Defn: (V-elliptic / Coercive) B is V-elliptic (aka Coercive) if there exists a $\beta_* > 0$ such that $B(v,v) \geq \beta_* \|v\|_V^2 \; \forall v \in V.$ (VE).

Defn: (Abstract Variational Problem) Let $B(\cdot,\cdot): V \times V \mapsto \mathbb{R}$ be a bilinear form that satisfies (S), (CB) and (VE). Let $L(\cdot): V \mapsto \mathbb{R}$ be a linear functional that is bounded in $\|\cdot\|_V$: $|L(v)| \leq \tilde{L}\|v\|_V \quad \forall v \in V.$

(1) Var. Prob: Find $u \in V$ s.t $B(u,v) = L(v) \;\forall\, v \in V.$

(2) Energy Minimization: Find $u \in V$ s.t $F(u) \leq F(w) \;\forall\, w \in V$, where

$$F(v) = \frac{1}{2} B(v,v) - L(v).$$

Claim: (1) & (2) are equivalent.

Proof: ($\Rightarrow$) Assume $u \in V$ satisfies Var. Prob. Let $w = u+v \in V$ for some $v \in V$.

$$F(w) = F(u+v) = \frac{1}{2} B(u+v, u+v) - L(u+v)$$

$$B(u+v,u+v) = B(u,u+v) + B(v,u+v) = B(u,u) + B(u,v) + B(v,u) + B(v,v)$$
$$= B(u,u) + 2B(u,v) + B(v,v) \quad (\text{Note: used (S)}).$$

$$\Rightarrow F(u+v) = \frac{1}{2} B(u,u) + B(u,v) + \frac{1}{2} B(v,v) - L(u) - L(v)$$
$$= \underbrace{\left[\frac{1}{2} B(u,u) - L(u)\right]}_{F(u)} + \underbrace{\left[B(u,v) - L(v)\right]}_{=0 \text{ (By Var. Prob)}} + \frac{1}{2} B(v,v)$$

$$F(w) = F(u) + \frac{1}{2} B(v,v), \text{ but } B(v,v) \geq \beta_* \|v\|_V^2 > 0 \;\forall\, v \neq 0$$
(Note: using (VE))

$$\therefore F(u) \leq F(w) \;\forall\, w \in V.$$

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($\Leftarrow$) Assume $u \in V$ satisfies $F(u) \leq F(w) \quad \forall\, w \in V$.

Let $g_v(\varepsilon) = F(u + \varepsilon v)$ for some ~~fixed~~ fixed, but arbitrary $v \in V$.

$$g_v(\varepsilon) = F(u) + \varepsilon B(u,v) - \varepsilon L(v) + \frac{1}{2}\varepsilon^2 B(v,v).$$

$$g_v'(\varepsilon) = B(u,v) - L(v) + \varepsilon B(v,v)$$

$$g_v'(\varepsilon = 0) = 0 \Rightarrow B(u,v) = L(v). \quad \blacksquare$$

2/oct/26

Abstract Var. Formulation.

Find $u \in V$ s.t ~~Find~~ ~~B~~ $B(u,v) = L(v) \quad \forall\, v \in V$

(S) $B(u,v) = B(v,u) \quad \forall\, u,v \in V.$

(CB) $B(u,v) \leq \beta^* \|u\|_V \|v\|_V \quad \forall\, u,v \in V.$

(∇E) $B(v,v) \geq \beta_* \|v\|_V^2 \quad \forall\, v \in V$

"Energy" $F(w) = \frac{1}{2} B(w,w) - L(w)$

Thm: (Lax-Milgram): Let V be a Hilbert space w/ norm $\|\cdot\|_V$. Let $B(\cdot,\cdot) : V \times V \to \mathbb{R}$ be a bilinear form that satisfies (S), (CB) & (VE). Let $L(\cdot) : V \to \mathbb{R}$ be a linear functional bounded in $\|\cdot\|_V$. Then there exists a unique soln to Var. Prob: Find $u \in V$ s.t $B(u,v) = L(v) \quad \forall\, v \in V$ with the property that $\|u\|_V \leq \frac{\tilde{L}}{\beta_*}$ where $\tilde{L}, \beta_*$ are real positive constants.

Proof: Since $B(\cdot,\cdot)$ is symmetric & V-elliptic it induces a scalar product in V.

$$[u,v] := B(u,v) \quad \forall\, u,v \in V.$$

(VE) $[v,v] = B(v,v) \geq \beta_* \|v\|_V^2$

(CB) $[v,v] = B(v,v) \leq \beta^* \|v\|_V^2$

$\therefore \boxed{\beta_* \|v\|_V^2 \leq [v,v] \leq \beta^* \|v\|_V^2}$

$\therefore$ the norms $\sqrt{[v,v]}$ and $\|v\|_V$ are equivalent.

$\therefore$ the linear functional $L(\cdot)$ is bounded in both $\|\cdot\|_V$ & $\sqrt{[\cdot,\cdot]}$.

By Riesz representation theorem, there exists a unique $u \in V$ such that $L(v) = [u,v] \ \forall \ v \in V$.

$\therefore$ There exists a unique $u \in V$ s.t $B(u,v) = L(v) \ \forall \ v \in V$.

For the solution bound:

$$\beta_* \|u\|_V^2 \leq B(u,u) = L(u) \leq \tilde{L} \|u\|_V$$

$$\Rightarrow \beta_* \|u\|_V \leq \tilde{L}$$

$$\therefore \|u\|_V \leq \frac{\tilde{L}}{\beta_*} \quad \blacksquare$$

Conforming FEMs (for abstract var. Prob):

Replace V by a finite dimensional subspace: $\underbrace{V_h \subset V}_{\text{conforming}}$

$$\boxed{\text{Find } u_h \in V_h \text{ s.t. } B(u,v) = L(v) \ \forall \ v \in V_h.}$$

Lax-Milgram still applies to this, $\therefore$ there exists a unique soln w/ the property that $\|u_h\|_V \leq \frac{\tilde{L}}{\beta_*}$

Let $\{\phi_i\}_{i=1}^M$ be a basis for V_h.

Let $v = \phi_j \quad j = 1, \cdots, M$. & Let $u_h = \sum_{i=1}^{M} \xi_i \phi_i$

$$\therefore \sum_{i=1}^{M} B(\phi_i, \phi_j)\, \xi_i = L(\phi_j) \quad \text{for } j = 1, 2, \ldots, M.$$

$\therefore$ Linear system: $\underline{\underline{A}}\, \underline{\xi} = \underline{b}$

$$A_{ij} = B(\phi_i, \phi_j)$$
$$\xi = (\xi_1, \xi_2, \ldots, \xi_M)$$
$$b_j = L(\phi_j)$$

Claim: Under assumptions (S) & (VE), $\underline{\underline{A}}$ is symmetric positive definite.

Proof: (1) Symmetry of $\underline{\underline{A}}$ follows directly from (S):

$$A_{ij} = B(\phi_i, \phi_j) = B(\phi_j, \phi_i) = A_{ji}$$

(2) Let $w = \sum_{i=1}^{M} w_i \phi_i \in V_h$

$$B(w, w) \geq \beta_* \|w\|_V^2$$

if $w \not\equiv 0$, then $B(w, w) > 0$.

$$B\left(\sum_{i=1}^{M} w_i \phi_i, w\right) = \sum_{i=1}^{M} w_i B(\phi_i, w) = \sum_{i=1}^{M} w_i B\left(\phi_i, \sum_{j=1}^{M} w_j \phi_j\right)$$

$$= \sum_{i=1}^{M} \sum_{j=1}^{M} w_i \underbrace{B(\phi_i, \phi_j)}_{A_{ij}} w_j = \underline{w}^T \underline{\underline{A}}\, \underline{w} > 0 \text{ by (VE)}$$

$\therefore \underline{\underline{A}}$ is pos. def.

Note: this means a conforming FEM for this abstract var. prob will generate a SPD linear system, indepedent of # of dimensions, poly order, shape of solution domain Ω, type of mesh, etc...

Thm: (Cea's Lemma) Let V be a Hilbert space w/ norm $\|\cdot\|_V$. B is a bilinear form that satisfies (S), (CB) & (VE). L is a linear bounded functional (bounded in $\|\cdot\|_V$).
Let $V_h \subset V$ be a finite dimensional subspace of V.

Let

(Var Prob) Find $u \in V$ s.t. $B(u,v) = L(v) \quad \forall\, v \in V$.

(FEM) Find $u_h \in V_h$ s.t $B(u_h, v) = L(v) \quad \forall\, v \in V_h$

then $\|u - u_h\|_V \leq \frac{\beta^*}{\beta_*} \|u - v\|_V \quad \forall\, v \in V_h$.

Proof: Subtract (Var. Prob) & (FEM):

$B(u,v) - B(u_h, v_h) = 0 \quad \forall\, v \in V_h$.

$\therefore \boxed{B(u - u_h, v) = 0 \ \forall\, v \in V_h}$ Galerkin Orthogonality.

Take:

$$B(u-u_h, u-u_h) = B(u-u_h, u-v+v-u_h) = B(u-u_h, u-v) + B(u-u_h, v-u_h)$$

0 (By Galerkin Orth

$$\Rightarrow \beta_* \|u-u_h\|_V^2 \overset{(VE)}{\leq} B(u-u_h, u-u_h) = B(u-u_h, u-v)$$

$$\leq \beta^* \|u-u_h\|_V \|u-v\|_V \quad (CB).$$

$$\therefore \boxed{\|u-u_h\|_V \leq \frac{\beta^*}{\beta_*} \|u-v\|_V}$$ ▨

Note: (Error estimate in $\|\cdot\|_V$)

take $v = \Pi_h u$ ← project $u \in V$ into V_h.

$$\therefore \|u-u_h\|_V \leq \frac{\beta^*}{\beta_*} \|u - \Pi_h u\|_V$$

$\|u - \Pi_h u\|_V$ is the interpolation as measured in $\|\cdot\|_V$.

if you can generate a sequence of meshes & associated bases as $h \to 0$ such that $\|u - \Pi_h u\|_V \to 0$, then we have convergence in $\|\cdot\|$

$\|u - \Pi_h u\|_V = O(h^p)$ ← check

Ex: $-u'' = f$ in $(0,1)$

$u(0) = u(1) = 0$

$I(0,1)$ $\quad V = H_0^1(I)$

$$\|v\|_V^2 = \int_0^1 [(v')^2 + v^2]\,dx = \|v'\|_{L^2(I)}^2 + \|v\|_{L^2(I)}^2$$

$$B(u,v) = (u', v') = \int_0^1 u'v'\,dx$$

$$L(v) = (f, v) = \int_0^1 fv\,dx$$

Check Conditions:

1) Symmetric : $(u', v') = (v', u')$

2) Continuous / Bounded : $|B(u,v)| \le \|u'\|_{L^2(I)} \|v'\|_{L^2(I)} \le \|v\|_V \|u\|_V$ ($\beta^* = 1$)

3) V-ellipticity: $B(u,v) = \|v'\|_{L^2(I)}^2 = \frac{1}{2}\left(\|v\|_{L^2(I)}^2 + \|v'\|_{L^2(I)}^2\right)$

(FEM) Find $u_h \in V_h$ s.t $B(u_h, v) = L(v)$ $\forall\, v \in V_h$. then

$$\|u - u_h\|_V \le \frac{\beta^*}{\beta_*} \|u - v\|_V \quad \forall\, v \in V_h.$$

Proof: subtract Var. Prob. & FEM

$B(u,v) - B(u_h, v) = 0 \quad \forall\, v \in V_h.$

$$\boxed{B(u - u_h, v) = 0 \quad \forall\, v \in V_h}$$

Galerkin Orthogonality

Take $B(u-u_h, u-u_h) = B(u-u_h, u-v+v-u_h) = B(u-u_h, u-v) + B(u-u_h, v-u_h)$

$= 0$ (by Galerkin orth)

(V-ellipticity)

$$\beta_* \|u-u_h\|_V^2 \le B(u-u_h, u-u_h)$$
$$= B(u-u_h, u-v)$$
$$\le \beta^* \|u-u_h\|_V \|u-v\|_V \quad (CB)$$

$$\Rightarrow \|u-u_h\|_V \le \frac{\beta^*}{\beta_*} \|u-v\|_V$$

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Note: (Error estimate in $\|\cdot\|_V$)

Take $v = \pi_h u$ ← project $u \in V$ into V_h

$$\therefore \|u - u_h\|_V \leq \frac{\beta^*}{\beta_*} \|u - \pi_h u\|_V$$

Point care: $\|v\|_{L^2(I)} \leq \|v'\|_{L^2(I)}$ for $v \in H_0^1(I)$

$$B(v,v) \geq \frac{1}{2}\left(\|v'\|^2_{L^2(I)} + \|v\|^2_{L^2(I)}\right)$$

$$\geq \frac{1}{2}\|v\|^2_V$$

↖ β_*

(4) L is continous:

$$|L(v)| \leq \|f\|_{L^2(I)} \|v\|_{L^2(I)} \leq \|f\|_{L^2(I)} \|v\|_V$$

$$\therefore \tilde{L} = \|f\|_{L^2(I)}$$

→ existence, uniqueness, boundedness.

Note: Cea's Lemma:

$$\|u - u_h\|_V \leq 2\|u - v\|_V \quad \forall\, v \in V_h. \leftarrow \text{interpolation error.}$$

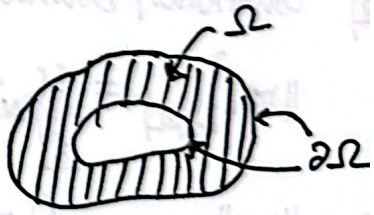
$$CG(1): \|u - u_h\|_V \leq c_1 \|h u''\|_{L^2(I)}$$

$$CG(2): \|u - u_h\|_V \leq c_2 \|h^2 u''\|_{L^2(I)}$$

Ex: 2D Poisson equation:

$$-\Delta u = -u_{,x,x} - u_{,y,y} = f \quad \text{on } \Omega \in \mathbb{R}^2$$

$$u = 0 \quad \text{on } \partial\Omega$$



Multiply by a test function v and integrate over Ω.

$$-\iint_\Omega v \Delta u \, dx\, dy = \iint_\Omega f v \, dx\, dy$$

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Note: $v\Delta u = v\,\underline{\nabla}\cdot(\underline{\nabla}u) = \nabla\cdot(v\,\underline{\nabla}u) - \underline{\nabla}u\cdot\underline{\nabla}v$ → Product rule.

$$-\iint_\Omega v\Delta u\,dx\,dy = \iint_\Omega \underline{\nabla}u\cdot\underline{\nabla}v\,dx\,dy + \iint_\Omega \underline{\nabla}\cdot(v\,\underline{\nabla}u)\,dx\,dy$$

By divergence theorem:

$$\iint_\Omega \underline{\nabla}\cdot(v\,\underline{\nabla}u)\,dx\,dy = \oint_{\partial\Omega} v\,\underline{\nabla}u\cdot\hat{\underline{n}}\,ds$$

$\hat{n}$: outward unit normal to $\partial\Omega$.

$$\Rightarrow \iint_\Omega \underline{\nabla}u\cdot\underline{\nabla}v\,dx\,dy + \int_{\partial\Omega} v\cdot\underline{\nabla}u\cdot\hat{\underline{n}}\,ds = \iint_\Omega fv\,dx\,dy$$

($\underline{\nabla}u\cdot\hat{\underline{n}} \to 0$)

Assume: $v=0$ on $\partial\Omega$

$$\therefore \iint_\Omega \underline{\nabla}u\cdot\underline{\nabla}v\,dx\,dy = \iint_\Omega fv\,dx\,dy$$

$$V = H_0^1(\Omega) = \left\{ v : \iint_\Omega (v^2 + \|\underline{\nabla}v\|^2)\,dx\,dy < \infty,\ v=0 \text{ on } \partial\Omega \right\}$$

$$\boxed{v^2 + \|\underline{\nabla}v\|^2 = v^2 + (v_{,x})^2 + (v_{,y})^2}$$

$$B(u,v) := \iint_\Omega \underline{\nabla}u\,\underline{\nabla}v\,dx\,dy \qquad L(v) := \iint_\Omega fv\,dx\,dy$$

(Var. Prob.): Find $u\in V$ s.t $B(u,v) = L(v)$ $\forall\ v\in V$.

Check Conditions:

1) Symmetric: $B(u,v) = B(v,u)$

2) Continuous / Bounded: $|B(u,v)| \le \|\nabla u\|_{L^2(\Omega)}\,\|\nabla v\|_{L^2(\Omega)}$

$$\|w\|^2_{L^2(\Omega)} = \iint_\Omega w^2\,dx\,dy$$

$$\|\nabla w\|_{L^2(\Omega)} = \iint_\Omega \nabla w\cdot\nabla v\,dx\,dy$$

Now, $\|w\|_V^2 = \|w\|^2_{H^1(\Omega)} = \|w\|^2_{L^2(\Omega)} + \|\nabla w\|^2_{L^2(\Omega)}$

$\Rightarrow |B(u,v)| \le \|u\|_V\,\|v\|_V$ Here, $\beta^* = 1$.

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(3) V-ellipticity:

$$B(v,v) = \|\nabla v\|^2_{L^2(\Omega)} = \frac{1}{2}\left(\|\nabla v\|^2_{L^2(\Omega)} + \|\nabla v\|^2_{L^2(\Omega)}\right)$$

Multi-D poincaré inequality: $\|v\|_{L^2(\Omega)} \leq C|v|_{H^1(\Omega)} = C\|\nabla v\|_{L^2(\Omega)}$ $\forall\, v \in H^1_0(\Omega)$

$$B(v,v) \geq \frac{1}{2}\left(\frac{1}{C^2}\|v\|^2_{L^2(\Omega)} + \|\nabla v\|^2_{L^2(\Omega)}\right)$$

$$B(v,v) \geq \frac{1}{2}\min\left(\frac{1}{C^2}, 1\right)\|v\|^2_V \rightarrow \beta^* = \frac{1}{2}\min\left(\frac{1}{C^2}, 1\right)$$

→ C is a constant depending only on Ω.

(4) L is continous:

$$|L(v)| \leq \|f\|_{L^2(\Omega)}\|v\|_{L^2(\Omega)} \leq \|f\|_{L^2(\Omega)}\|v\|_V.$$

$$\tilde{L} = \|f\|_{L^2(\Omega)}$$

9-Oct-2026

ex: 2D-Poisson

$$\left\{\begin{array}{l} -\Delta u = f \text{ in } \Omega \\ u = 0 \text{ on } \partial\Omega \end{array}\right\}$$

$V = H^1_0(\Omega)$

CG(1) FEM (more details later)

Mesh T_h: subdivide Ω into a finite # of triangles

K_j for $j = 1, 2, \ldots, M$

$h_j :=$ longest edge of K_j

$h := \max_{1 \leq j \leq m} h_j$

$\rho_j :=$ diameter of the largest circle incribed in K_j

Def: T_h is regular if there exists a constant $\alpha > 0$ such that

$$\frac{h_j}{\rho_j} \leq \alpha \quad \forall j,$$ where α is independent of h.

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ex: $V_h = \{ v \in H^1(\Omega) : v|_{K_j} = a_j + b_j x + c_j y,\ v|_{\partial\Omega} = 0 \}$

its fomble $\partial\Omega$ is not the same as $\partial\tilde{\Omega}$.

Error estimate (Cea's lemma)

$$\|u - u_h\|_{H^1(\Omega)} \leq c \|hu\|_{H^2(\Omega)}$$

$$\|v\|^2_{H^2(\Omega)} = \iint_\Omega [(v,_{xx})^2 + 2(v,_{xy})^2 + (v,_{yy})^2]\, dx dy$$

ex: $\left\{ \begin{array}{l} u^{(4)} = f \quad \text{in } (0,1) \\ u(0) = u(1) = u'(0) = u'(1) = 0 \end{array} \right\}$

$V = H_0^2(I)$; $I = (0,1)$

$$\|v\|_V = \|v\|_{H^2(I)} = \int_0^1 [v^2 + (v')^2 + (v'')^2]\, dx$$

$H_0^2(I) \rightarrow u = 0$ on boundary
$u' = 0$ on boundary

$B(u,v) = (u'', v'')$

$L(v) = (f, v)$

(1) Symmetric : $(u'', v'') = (v'', u'')$

(2) Continous / Bounded : $|B(u,v)| \leq \|u''\|_{L^2(I)} \cdot \|v''\|_{L^2(I)}$
$\leq \|u\|_V \|v\|_V$
$\beta^* = 1$

(3) V-elliptic : $B(v,v) = \|v''\|^2_{L^2(I)}$

$$B(v,v) = \frac{1}{3}\left(\|v''\|^2_{L^2} + \|v''\|^2_{L^2} + \|v''\|^2_{L^2}\right)$$

Poincare $\rightarrow$ $\|v\|_{L^2(0,1)} \leq \|v'\|_{L^2(0,1)}$
$\leq \|v''\|_{L^2(0,1)}$ $\forall\ v \in H_0^2(0,1)$

$$\therefore B(v,v) \geq \frac{1}{3}\left(\|v\|_{L^2}^2 + \|v'\|_{L^2}^2 + \|v''\|_{L^2}^2\right)$$

$$B(v,v) \geq \frac{1}{3}\|v\|_V^2$$

$$\beta^* = \frac{1}{3} \quad \checkmark$$

(4) L is continous : $\tilde{L} = \|f\|_{L^2(0,1)}$

cG(3) estimate: Cea's lemma:

$$\|u - u_h\|_{H^2(0,1)} \leq C |h^2 u|_{H^4(0,1)}$$

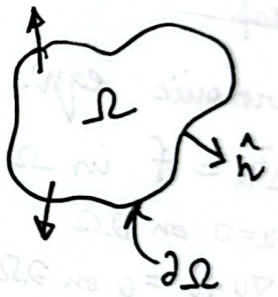
$$|v|_{H^4(0,1)}^2 = \int_0^1 (v'''')^2 dx$$

ex: Biharmonic equation:

$$\Delta^2 u \Rightarrow \Delta\Delta u = f \rightarrow \text{in } \Omega$$

$$\Rightarrow u_{,x,x,x,x} + 2u_{,x,x,y,y} + u_{,y,y,y,y} = f$$

B.Cs: $u|_{\partial\Omega} = 0$, $\nabla u \cdot \hat{n}|_{\partial\Omega} = 0$



Multiply by V and integrate :

$$\iint_\Omega v \Delta\Delta u \, dx dy = \iint_\Omega v f \, dx dy$$

Let $z = \Delta u$

$$\Rightarrow \iint v \Delta z \, dx dy = \iint_\Omega v \nabla\nabla z \, dx dy$$

$$\Rightarrow -\iint_\Omega \nabla v \cdot \nabla z \, dx dy + \oint_{\partial\Omega} v \nabla z \cdot \hat{n} \, ds$$

take $v = 0$ on $\partial\Omega$,

$$-\iint_\Omega \underline{\nabla} v \cdot \underline{\nabla}(\Delta u) dx dy = \iint_\Omega (\underline{\nabla} \cdot \underline{\nabla} v) \Delta u \, dx dy - \oint_{\partial\Omega} \Delta u \nabla v \cdot \hat{n} \, ds$$

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take $\nabla v \cdot \hat{n} = 0$ on $\partial\Omega$

$$B(u,v) = \iint \Delta u \cdot \Delta v \, dx dy$$

$$L(v) = \iint_\Omega f v \, dx dy$$

$$V = H_0^2(\Omega)$$

Note: if $v \in H_0^2(\Omega)$

then $\|v\|^2_{H^2(\Omega)} = \|v\|^2_{L^2(\Omega)} + \|\nabla v\|^2_{L^2(\Omega)} + \|v_{,xx}\|^2_{L^2(\Omega)}$

$$+ 2\|v_{,xy}\|^2_{L^2(\Omega)} + \|v_{,yy}\|^2_{L^2(\Omega)}$$

$v \in H_0^2(\Omega)$

$\hookrightarrow v = 0$ on $\partial\Omega$

$v_{,x} = 0$ on $\partial\Omega$

$v_{,y} = 0$ on $\partial\Omega$

12-Oct-26 Mon Recap

ex: Biharmonic eqn:

$$\begin{cases} \Delta^2 u = f \text{ in } \Omega \\ u = 0 \text{ on } \partial\Omega \\ \frac{\partial u}{\partial n} \cdot \hat{n} = 0 \text{ on } \partial\Omega \end{cases}$$

$V = H_0^2(\Omega)$

$B(u,v) = (\Delta u, \Delta v)$

$L(v) = (f, v)$

$F(w) = \frac{1}{2} B(w,w) - L(w)$

Check Conditions:

(1) Symmetric: $B(u,v) = B(v,u)$

(2) Continous/Bounded: $|B(u,v)| \leq \|\Delta u\|_{L^2(\Omega)} \|\Delta v\|_{L^2(\Omega)}$

$$\leq \|u\|_V \|v\|_V$$

(3) V-ellipticity: $B(v,v) = \|\Delta v\|^2_{L^2(\Omega)}$

(Girault-Raviart, 1981):

$$\|v\|_{H^2(\Omega)} \leq c\|\Delta v\|_{L^2(\Omega)}$$

for $v \in H^2_0(\Omega)$

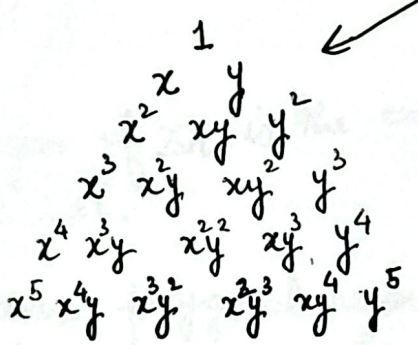
$$\therefore B(v,v) \geq \frac{1}{c^2}\|v\|^2_V$$

(4) $\tilde{L} = \|f\|_{L^2(\Omega)}$.

Conforming FEM:

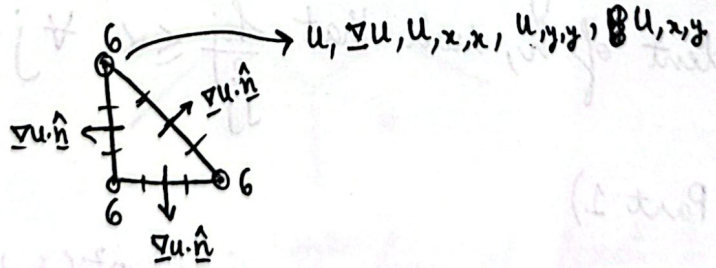
$$V_h \subset V = H^2_0(\Omega)$$

$$V_h = \left\{ v \in H^2(\Omega) : \; v|_{K_j} = \sum_{l,m=0}^{m+l=5} a_j^{(l,m)} x^{(l)} y^{(m)}, \; v|_{\partial\Omega} = 0, \; \nabla u \cdot \hat{\underline{n}}\big|_{\partial\Omega} = 0 \right\}$$



(21) different polynomials.

Argyris element (1968):



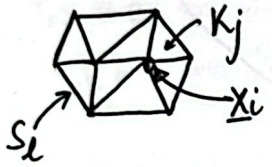
2D Mesh Generation: (triangles)

Note: FEM have no ~~interest~~ inherent restrictions on the mesh. Simplest and most flexible is to use a simplex: 2D triangles, 3D tetrahedrons

Note: a general triangulation in 2D is unstructured, meaning grid points don't lie along coordinate lines.

Defn: Let Ω be a 2D domain w/ polygonal boundary $\partial\Omega$. A triangulation, $T_h = \{K_j\}_{j=1}^{M}$, is a subdivision of Ω into M non-overlapping triangles constructed so that no vertex of one triangle lies on the edge of another triangle: $\bigcup_{j=1}^{M} T_j = \Omega$

Let $X_h = \{\underline{x}_i = (x_i, y_i)\}_{i=1}^{N}$ denote the set of nodes (vertices of the triangles). Let $S_h = \{S_\ell\}_{\ell=1}^{E}$ be the edges of the triangles.



h_j = longest edge of K_j.

$h = \max_{1 \le j \le M} h_j$

ρ_j = diameter of the largest circle inscribed in K_j.

Defn: T_h is regular triangulation if there exists a constant $\alpha > 0$, independent of h, such that $\frac{h_j}{\rho_j} \le \alpha \ \forall j$.

Mesh Generation (Part 1)

Main Question: Given a collection of points in $\mathbb{R}^2$ (nodes), how can we connect them to form a triangulation?

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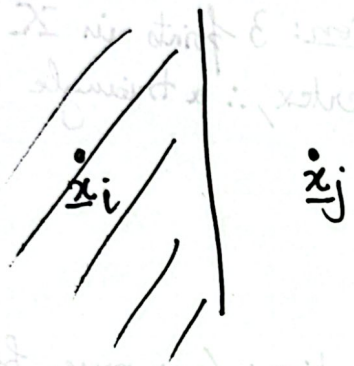
Idea: given a finite set of points in $\mathbb{R}^2$, assign to each point a region of influence in such a way that it decomposes $\mathbb{R}^2$.

Defn: (Voronoi Region) Let X be a collection of N points in $\mathbb{R}^2$. For each $\underline{x}_i \in X$ the **Voronoi Region**, V_i, is the set of points $\underline{x} \in \mathbb{R}^2$ that are at least as close to $\underline{x}_i$ as any other point in X.

$$V_i = \{\underline{x} \in \mathbb{R}^2 : \|\underline{x} - \underline{x}_i\| \leq \|\underline{x} - \underline{x}_j\|, \ \forall \ \underline{x}_j \in X\}$$

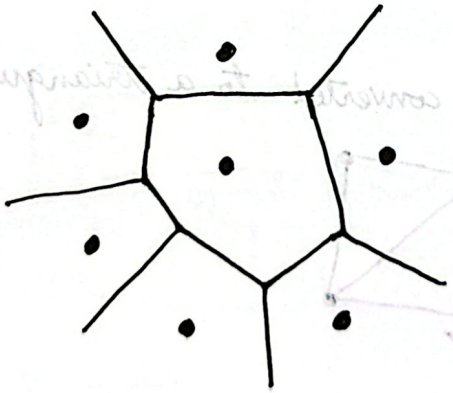
Consider the half-plane:

$$H_{ij} = \{\underline{x} \in \mathbb{R}^2 : \|\underline{x} - \underline{x}_i\| \leq \|\underline{x} - \underline{x}_j\|\}$$



∴ Voronoi region of $\underline{x}_i$ is the ~~convex~~ intersection of the half-planes H_{ij} for all $j \neq i$.

∴ V_i is a convex polygonal region (possibly unbounded).



Note: (1) Each point in $\mathbb{R}^2$ has at least 1 nearest point in X,

∴ every $\underline{x} \in \mathbb{R}^2$ will lie in at least 1 Voronoi region.

(2) 2 Voronoi regions lie on opposite sides of a ⊥ bisector, 2 Voron region don't share any interior points. If a point belongs to more than

1 Voronoi region, it must be on the boundary.

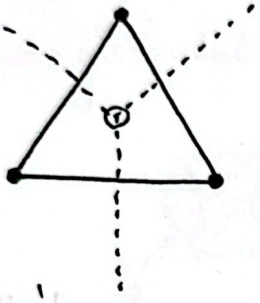
(3) Voronoi regions for a given X cover $\mathbb{R}^2$.

(4) Voronoi regions together with edges and vertices form the Voronoi Diagram of X.

(5) For a given X, the Voronoi diagram is unique.

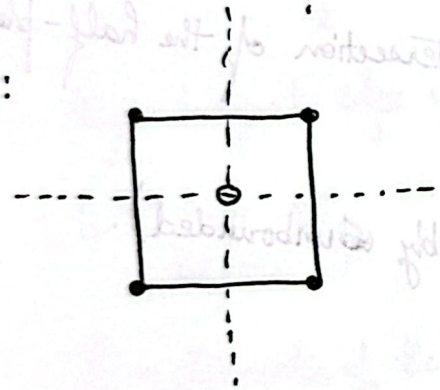
Defn: (Delaunay Partition) Let $X \in \mathbb{R}^2$ be a set of N points. Let V be the Voronoi diagram. The Delaunay partition is obtained by creating an edge between any 2 points in X, if and only if their Voronoi regions share a boundary.

ex:



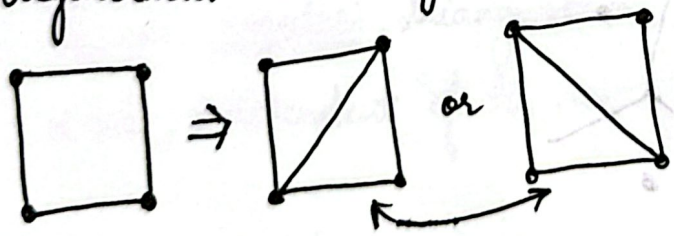
Generic situation: 3 points in X share a common vertex, ∴ a triangle is formed.

ex:



Degenerate situation: 4 or more points share a common Voronoi vertex if they are co-circular.

Note: degeneracies can always be converted to a triangulation.



Defn: (Delaunay Triangulation) The Delaunay Triangulation, T_h, is the Delaunay Partition with some strategy to convert degenerate elements into triangles.

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Note: (1) Voronoi diagram & Delaunay triangulation form a duality.

(2) For a given $X \in \mathbb{R}^2$, V is unique, T_h is unique up to degeneracies.

(3) T_h contain $O(N)$ triangles.

(4) T_h maximizes the minimum angle.

(5) A circle circumscribing any triangle in T_h does not contain any other point in X.

- In Python:

1) Create a list of points in $\mathbb{R}^2$: `p[0:N, 0:2]` (point index; coordinate index: (x,y))

2) Create T_h (Delaunay Triangulation)

```
import scipy.spatial as sp
t = sp.Delaunay(p).simplices
```

3) Plot mesh:

```
import numpy as np
import matplotlib.pyplot as plt
plt.tricolor[p[:,0], p[:,1], t, facecolors = np.zeros(N), edgecolors = 'k', cmap =
```

4) Plot function defined on the nodes:

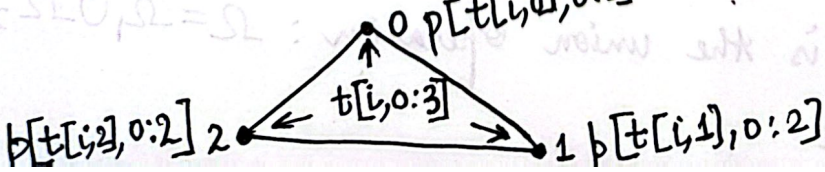
```
U = ufunc(p)
plt.tripcolor(p[:,0], p[:,1], t, U, shading = 'flat', cmap = 'inferno')
```

→ What is t?

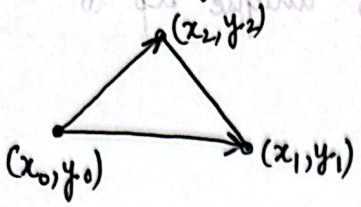
N = # of nodes

M = # of triangles

t is an Mx3 array of integers, where t[i,j]: i triangle, j node inde (0,1,2



Note: area of a triangle



$\underline{v}_1 = (x_2 - x_0, y_2 - y_0, 0)$

$\underline{v}_2 = (x_1 - x_0, y_1 - y_0, 0)$

$\text{area} = \frac{1}{2} \left| \hat{\underline{k}} \cdot (\underline{v}_1 \times \underline{v}_2) \right|$

Note: Delaunay in C: http://www.qhull.org

⇒ **Part 2**: (how to represent Ω and $\partial\Omega$?)

Idea: Describe the domain *implicitly* using a level set function.

ex: $\phi(\underline{x}) = \|\underline{x}\| - R$, where $R > 0$.

$$\partial\Omega = \{ x \in \mathbb{R}^2 : \phi(\underline{x}) = 0 \}$$

$$= \{ \underline{x} \in \mathbb{R}^2 : \underbrace{\|\underline{x}\| = R}_{x^2 + y^2 = R^2} \}$$

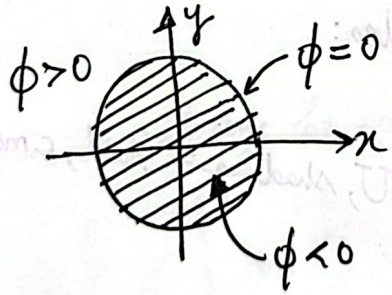
↳ circle of radius R centered at $(0,0)$.

$$\Omega = \{ \underline{x} : \mathbb{R}^2 : \phi(\underline{x}) \leq 0 \}$$

Note: by convention: $\{\phi = 0\}$ is boundary

$\{\phi < 0\}$ is interior of Ω

$\{\phi > 0\}$ is exterior of Ω.



Note: for a given $\partial\Omega$ there are many level sets functions:

e.g, $\phi(\underline{x}) = \|\underline{x}\|^p - R^p$ for any positive integer p, describe the disc of radius R centered at $(0,0)$.

Boolean operations:

1) $\phi_{\Omega} = \min(\phi_1, \phi_2)$ is the union operation: $\Omega = \Omega_1 \cup \Omega_2$

(ϕ_{Ω} ↑ Ω; ϕ_1 ↑ Ω_1; ϕ_2 ↑ Ω_2)

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2) $\phi = \max(\phi_1, \phi_2)$ is the intersection operation $\Omega = \Omega_1 \cap \Omega_2$.

ϕ : Ω ; ϕ_1 : Ω_1 ; ϕ_2 : Ω_2

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